

Technology #04

Passive Drainage Priority Safety Floor

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

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Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Passive Drainage Priority Safety Floor

WHY DOES IT MATTER?

Obstruction causes 30-50% of shunt failures with no passive safety floor

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 4 governing equations, 10 design inputs, 5 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in Hydraulic Passive Safety + Multi-Lumen Valve Mechanics. 5-step build plan with specific test articles and acceptance criteria. Target buyer: Neuroshunt manufacturer; catheter engineering company.

WHAT DOES THE BUYER GET?

Passive floor concept + mechanism description; Parallel conductance governing equation; Bench test protocol (obstruction, kink, Cpk); Critical common-cause obstruction UNKNOWN disclosure; This engineering dossier with failure analysis + build plan

WHAT DOES THE BUYER HAVE TO BUILD?

Production multi-lumen catheter design (dimensioned, toleranced); Multi-lumen extrusion process + vendor qualification; Common-cause obstruction validation evidence; Regulatory submission (510(k) or De Novo); Clinical validation (revision rate endpoint); Manufacturing scale-up

WHAT IS THE NEXT DECISIVE EXPERIMENT?

Multi-lumen extruded samples (various $d_{\text{floor}}/d_{\text{primary}}$ ratios)

WHAT WOULD MAKE US KILL IT?

Common-cause obstruction affects both primary and floor lumens equally (no differential immunity)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 5 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 5-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: Multi-lumen extruded samples (various $d_{\text{floor}}/d_{\text{primary}}$ ratios)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - Multi-lumen extruded samples (various $d_{\text{floor}}/d_{\text{primary}}$ ratios) - 10 weeks

Basis: Build plan WP. Scope: G_{floor} , G_{primary} , dimensional accuracy. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - Kink test specimens - 4 weeks

Basis: Build plan WP. Scope: Flow vs bend radius. Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - Biocompatibility specimens - 12 weeks (external lab)

Basis: Build plan WP. Scope: Standard ISO 10993 series. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 4 equations. Key assumptions: Both paths laminar ($Re < 2300$); Paths are hydraulically independent (no cross-talk); Outlet pressure identical for both paths; Floor path is immune to the same obstruction mechanism (CRITICAL ASSUMPTION — may not hold). UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

5 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: Multi-lumen extruded samples (various $d_{\text{floor}}/d_{\text{primary}}$ ratios). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Multi-lumen parallel catheter with differential conductance maintaining minimum drainage when primary path obstructs

POTENTIAL VALUE DRIVER: Addresses: Obstruction causes 30-50% of shunt failures with no passive safety floor

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: Neuroshunt manufacturer; catheter engineering company

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: Hydraulic Passive Safety + Multi-Lumen Valve Mechanics

Disciplines: Hydraulic conductance engineering, Multi-lumen catheter design, Failure mode analysis (FMEA), Polymer extrusion

Architecture: CSF shunt with passive safety floor: a secondary drainage path with lower conductance that maintains minimum drainage when the primary path obstructs. Differential pressure thresholds activate the floor path.

Subsystems:

SS-01: Primary drainage lumen - Primary CSF drainage path with higher conductance G_{primary}

SS-02: Secondary (safety floor) lumen - Lower-conductance path that opens when primary obstructs (pressure threshold or mechanical backup)

SS-03: Pressure-activated switch (optional) - Mechanical switch that activates floor path when primary path dP exceeds threshold

SS-04: Outlet catheter (common) - Drains CSF to peritoneal/atrial/pleural destination

2. Mechanism Architecture

Physical Changes: When primary lumen obstructs (debris, tissue ingrowth, protein aggregation), pressure upstream rises; secondary lumen either passively continues to drain (always open, lower conductance) or opens via pressure-activated switch.

Key Physics: Parallel hydraulic conductance: $G_{\text{total}} = G_{\text{primary}} + G_{\text{floor}}$. When $G_{\text{primary}} \rightarrow 0$ (obstruction), $G_{\text{total}} \rightarrow G_{\text{floor}}$. Floor conductance sized to maintain minimum drainage $Q_{\text{min}} = G_{\text{floor}} * dP$.

3. Governing Engineering Model

Summary: Parallel hydraulic conductance model. Two drainage paths in parallel: primary (high conductance) and floor (lower conductance). When primary obstructs, floor maintains minimum drainage.

Equations:

$Q = (\pi * r^4 * dP) / (8 * \eta * L)$ [P-07 multi-lumen floor conductance – laminar flow per lumen]

$Q_{\text{total}} = Q_{\text{primary}} + Q_{\text{floor}}$

$G_{\text{total}} = G_{\text{primary}} + G_{\text{floor}}$ [parallel conductance] [P-07 parallel floor conductance – passive safety floor mechanism]

$Q_{\text{floor_min}} = G_{\text{floor}} * dP_{\text{max}}$ [minimum drainage at worst-case pressure]

Assumptions:

- Both paths laminar ($Re < 2300$)
- Paths are hydraulically independent (no cross-talk)
- Outlet pressure identical for both paths
- Floor path is immune to the same obstruction mechanism (CRITICAL ASSUMPTION — may not hold)

Boundary Conditions:

- Inlet: ICP 5-40 mmHg
- Outlet: peritoneal pressure ~5-15 mmHg
- dP across shunt: 0-35 mmHg

Failure Regimes:

- Floor lumen obstructs by same mechanism as primary (common-cause failure)
- G_{floor} too high $\rightarrow$ over-drainage when primary patent
- G_{floor} too low $\rightarrow$ under-drainage when primary obstructed
- Floor lumen too small $\rightarrow$ manufacturing tolerance dominates

4. Design Inputs

ID	Input	Value	Origin
DI-001	Clinical need	Obstruction causes 30-50% of shunt failures; no passive safety floor exists (R332)	REDEFINED
DI-002	Functional requirement	Passive safety mechanism maintaining minimum drainage when primary path is obstructed	MODELLED
DI-003	Flow regime	Laminar, Re << 2300 (multi-lumen parallel flow; per-lumen Re << 2300)	COMPUTATIONALLY_SUPPORTED
DI-004	ICP range	ICP 5-40 mmHg (floor activation threshold range; passive safety engages at VIGNIER)	VERIFIED
DI-005	Q_min target (minimum drainage)	UNKNOWN — MODEL_DERIVED candidate ~0.05 mL/min (per R332)	MODELLED
DI-006	Q_max target (over-drainage limit)	UNKNOWN	UNKNOWN
DI-007	Biocompatibility (ISO 10993)	UNKNOWN	UNKNOWN
DI-008	Sterilization	UNKNOWN — likely EtO or gamma	UNKNOWN
DI-009	Multi-lumen extrusion tolerance	UNKNOWN	UNKNOWN
DI-010	Catheter body material	UNKNOWN — likely silicone or polyurethane (external precedent)	EXTERNAL_PRECEDENT

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	Multi-lumen catheter geometry (primary + floor lumen channels, length, layout)	ABSENT	G_floor target; G_primary target; Multi-lumen layout
DO-002	Extrusion process specification	ABSENT	Vendor selection; Tolerance requirements; Material selection frozen
DO-003	Bench test protocol for obstruction	CONCEPTUAL	

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
G_floor (floor conductance)	UNKNOWN	mL/(min·mmHg)	Design choice balancing Q_min and over-drainage	Bench flow measurement
G_primary (primary conductance)	UNKNOWN	mL/(min·mmHg)	Standard shunt catheter geometry	Bench flow measurement
Floor lumen diameter d_floor	UNKNOWN	mm	Derived from G_floor target via Hagen-Poiseuille	Agenda inspection
Q_min (minimum drainage)	UNKNOWN	mL/min	Clinical need (prevents over-drainage)	Calibration/re review
Obstruction resistance threshold	UNKNOWN	mmHg/(mL/min)	Pressure at which floor dominates	Obstruction bench test

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
Common-cause obstruction	Same debris/tissue that obstructs primary also obstructs floor	Different lumen geometry, standardise obstruction	Comparative obstruction	UNKNOWN whether floor is genuinely independent
Over-drainage when primary patchy	Primary patchy high relative to G_floor	G_floor ratio to maintain Q_total	Bench flow safety	UNKNOWN optimal ratio
Under-drainage when primary obstructed	Primary obstructed	Increase d_floor	Obstruction bench test	UNKNOWN optimal G_floor
Manufacturing tolerance violation	Extrusion too small -> tolerance	Process capability study (Cp, Cpk)	Cpk measurement	UNKNOWN Cpk for multi-lumen extrusion
Floor lumen collapse (kink)	Mechanical deformation close to lumen	Reinforcement or larger diameter	Kink test per ISO 10555-1	UNKNOWN kink resistance of multi-lumen design

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
Common-cause obstruction	Surface occlusion mechanism affects both lumen	Differing geometry, material, or surface treatment	CRITICAL UNKNOWN — may invalidate concept
Over-drainage when primary patent	Q_floor too high	Tune ratio	
Under-drainage when primary obstructed	Q_floor too low	Increase d_floor	
Manufacturing tolerance violation	Tolerance > margin	Cpk study + tolerance stack analysis	
Kink-induced occlusion	Mechanical deformation closes lumen	Reinforcement or larger diameter	

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
V-001	Q_floor >= Q_min when primary obstructed	Bench obstruction test	Q_floor >= Q_min target	NOT_TESTED
V-002	Q_total within bounds when primary patent	Bench flow test	Q_total within safety range	NOT_TESTED
V-003	Process capability Cpk >= 1.33	Cpk study	Cpk >= 1.33	NOT_TESTED
V-004	Kink resistance per ISO 10555-1	ISO 10555-1 kink test	Pass	NOT_TESTED

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VAL-001	Reduced revision rate vs standard shunt	Clinical trial	Statistically significant reduction	NOT_PERFORMED

11. Materials

Component	Candidate Material	Source	Verification Required	Status
Multi-lumen catheter body	Silicone elastomer (radiopaque)	Standard CSF shunt material	ISO 10993 + multi-lumen extrusion	CANDIDATE
Multi-lumen catheter body	Polyurethane (radiopaque)	Some shunts use polyurethane	ISO 10993 + multi-lumen extrusion	CANDIDATE

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
01	Multi-lumen catheter body	1	CUSTOM_COMPONENT	UNKNOWN — silicone or polyurethane (BIOCOMPATIBLE)	UNKNOWN	CRITICAL	Dimensional + flow + kink
02	Inlet (ventricular) catheter connector	1	CUSTOM_COMPONENT	UNKNOWN — likely radiopaque polymer	Multiple	HIGH	Dimensional + biocompatibility
03	Outlet (peritoneal) catheter	1	CUSTOM_COMPONENT	UNKNOWN	Multiple	HIGH	Dimensional + biocompatibility

13. Manufacturing

Status: ENGINEERING_CANDIDATE — no multi-lumen floor process validated for this specific design

Candidate Processes:

- Multi-lumen extrusion: Standard medical tubing process (multiple lumens achievable) - Requires custom extrusion die; vendor capability varies
- Radiopaque compound compounding: Barium sulfate or bismuth trioxide loading - Affects mechanical properties

14. External Evidence

Source 1: PurSafe Passive Safety IV Catheter - YouTube

URL: <https://www.youtube.com/watch?v=ZrEt0fs0cEc>

Excerpt: Nov 29, 2024 · Featuring: A passive fail-safe safety clip compliant with NSPA standard The MedSource triple-facet back-cut needle tip An optimized catheter ...

Source 2: A prospective randomized trial of two safety peripheral intravenous

URL: <https://pubmed.ncbi.nlm.nih.gov/18635482>

Excerpt: Needle withdrawal was more difficult with the passive safety catheter. Safety catheters are not superior with regard to failure rate in the catheter's ...

Source 3: Catheter & Drain Care Safety focuses on the safe handling, maintenance

URL: <https://www.instagram.com/reel/DWODBMJj5XP>

Excerpt: Mar 23, 2026 · Catheter and drain care safety focuses on safe handling, maintenance, and continuous monitoring of urinary catheters and surgical drains to ...

Source 4: Indwelling Urinary Catheter-related Problems - UroToday

URL: <https://www.urotoday.com/library-resources/indwelling-catheters/133482-indwelling-urinary-catheter-related-problems.html>

Excerpt: In this study, CAUTIs were marginally associated with catheter blockage but other catheter-related problems reported included blockage (34%), accidental ...

Source 5: Urinary catheter management - Starship Hospital

URL: <https://www.starship.org.nz/guidelines/urinary-catheter-management>

Excerpt: Feb 19, 2026 · Over inflation of the balloon can cause occlusion of drainage eyes, irritate the bladder wall, and lead to bladder spasms, patient discomfort ...

15. Transfer Boundary

YOU RECEIVE:

- + Passive floor concept + mechanism description
- + Parallel conductance governing equation
- + Bench test protocol (obstruction, kink, Cpk)
- + Critical common-cause obstruction UNKNOWN disclosure
- + This engineering dossier with failure analysis + build plan

YOU MUST DEVELOP:

- > Production multi-lumen catheter design (dimensioned, toleranced)
- > Multi-lumen extrusion process + vendor qualification
- > Common-cause obstruction validation evidence
- > Regulatory submission (510(k) or De Novo)
- > Clinical validation (revision rate endpoint)
- > Manufacturing scale-up

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- Achievable G_{floor} / G_{primary} ratio given manufacturing constraints — UNKNOWN
- Whether floor is genuinely immune to common-cause obstruction — UNKNOWN (critical)
- Optimal Q_{min} for clinical safety — UNKNOWN (clinical literature gap)
- Multi-lumen extrusion process capability — UNKNOWN
- Regulatory pathway (510(k) with substantial equivalence to existing shunts, or De Novo if no predicate for multi-lumen floor) — UNKNOWN
- Clinical benefit magnitude for passive safety floor — reduction in obstruction-related revision rate vs standard shunt — UNKNOWN until trial

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	Multi-lumen extruded samples (various G_{floor} / G_{primary} ratios)	ISO 10555-1 kink test rig	Flow vs bend radius	No occlusion at specified bend radius	ISO 10555-1 kink test report	4 weeks
WP-02	Bench obstruction rig	Obstruction simulator, flow vs pressure	Flow vs pressure	No occlusion at specified pressure	Obstruction test report	6 weeks
WP-03	Kink test specimens	ISO 10555-1 kink test rig	Flow vs bend radius	No occlusion at specified bend radius	ISO 10555-1 kink test report	4 weeks
WP-04	Comparative obstruction test specimens (primary vs floor)	Obstruction simulator, flow vs pressure	Flow vs pressure	No occlusion at specified pressure	Obstruction test report	6 weeks
WP-05	Biocompatibility specimens	ISO 10993 test lab	Standard ISO 10993 tests	ISO 10993 pass	ISO 10993 report	12 weeks (external lab)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 4 governing equations and a 5-step build plan. The next experiment is clearly specified. Target buyer: Neuroshunt manufacturer; catheter engineering company.

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: Common-cause obstruction affects both primary and floor lumens equally (no differential immunity)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: Common-cause obstruction affects both primary and floor lumens equally (no differential immunity)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.